Notes

Introduction to JEE Advanced Preparation

Welcome and Overview

Hello everyone! This session is dedicated to preparing for JEE Advanced. We understand that JEE Advanced is approaching, and our goal is to help you achieve a very good percentile in JEE Mains and qualify for JEE Advanced.

Chemistry Chapters Covered

I will be covering the following chemistry topics:

- Coordination Compounds
- General Organic Chemistry (GOC)
- Other main organic chemistry parts

Other parts of chemistry will be covered by another instructor. You may have already started other chemistry classes, possibly covering topics like atomic structure or physical chemistry chapters such as thermodynamics. Thermodynamics is also a very important chapter for JEE.

Introduction to Basic Chemistry and Chemical Bonding

Importance of Basic Chemistry

Basic chemistry, particularly the chapter on chemical bonding, is fundamental for understanding all other chemistry topics. It is a crucial area for both JEE Mains and JEE Advanced examinations, with numerous questions originating from these concepts.

Prerequisite for Coordination Compounds

Before delving into coordination compounds, it is essential to have a strong understanding of how bonding occurs, how bonds are formed, and the concept of hybridization. A firm grasp of these basic topics is necessary to comprehend the complexities of coordination compounds. This foundational knowledge ensures a thorough understanding rather than a superficial one.

Types of Chemical Bonds

Introduction to Chemical Bonds

Chemical bonds are broadly classified into two main types:

- **Ionic Bond:** Formed by the transfer of electrons.
- **Covalent Bond:** Formed by the sharing of electrons.

element to an electronegative element.

- **Electropositive Element:** Loses electrons.
- **Electronegative Element:** Gains electrons.

Factors Affecting Ionic Bond Formation

The tendency to form an ionic bond primarily depends on the electronegativity difference between the bonding atoms. Let's consider two atoms, A and B, forming an ionic bond.

Electronegativity Difference (ΔEN)

Electronegativity difference (ΔEN) is a crucial factor in determining the nature of a chemical bond.

- If $\Delta EN > 1.6$, the bond is predominantly ionic.
- If $\Delta EN < 1.6$, the bond is predominantly covalent.
- If $\Delta EN = 1.6$, the bond has 50% ionic character (I.C.) and 50% covalent character (C.C.).

Purity of Bonds

It is important to note that a bond between different atoms can never be purely ionic or purely covalent. Every ionic bond will possess some degree of covalent character. However, a pure covalent bond can exist between identical atoms (e.g., in diatomic molecules like H_2, O_2).

Introduction to Covalent Bonds and Explanatory Theories

Covalent Bond Formation

A covalent bond is formed by the sharing of electrons between atoms. This is a fundamental concept in chemistry.

Theories Explaining Covalent Bonding

Several theories have been developed to explain the nature and properties of covalent bonds. These include:

- **Valence Bond Theory (VBT):** This was one of the earliest theories to explain covalent bonding. It focuses on the overlapping of atomic orbitals and hybridization.
- **Molecular Orbital Theory (MOT):** This theory describes the formation of molecular orbitals from atomic orbitals.
- **VSEPR Theory (Valence Shell Electron Pair Repulsion Theory):** This theory primarily explains the shapes of molecules based on the repulsion between electron pairs around a central atom.

formation and molecular geometry.

- **VSEPR Theory:** Does not consider overlapping or hybridization. Its main focus is on the repulsions between electron pairs (both lone pairs and bond pairs) around the central atom, which dictates the molecular geometry.

VSEPR Theory: Electron Pair Repulsions

Understanding VSEPR Theory through Water (H_2O)

Let's consider the water molecule (H_2O) to understand the principles of VSEPR theory.

1. **Central Atom Identification:** In H_2O, oxygen is the central atom. Although the central atom is typically the least electronegative, hydrogen cannot be the central atom because it forms only one bond. Oxygen, being able to form multiple bonds, serves as the central atom.
2. **Bond Pairs and Lone Pairs:**
   - Oxygen forms two bonds with two hydrogen atoms, resulting in **two bond pairs**.
   - Oxygen has 6 valence electrons. Two are used in bonding, leaving 4 non-bonding electrons. These form **two lone pairs** on the oxygen atom.

Electron Pair Repulsions

All electron pairs (both lone pairs and bond pairs) around the central atom repel each other. This repulsion dictates the molecular geometry.

Order of Repulsions

The strength of repulsion between different types of electron pairs follows a specific order:

1. **Lone Pair - Lone Pair (LP-LP) Repulsions:** These are the strongest repulsions.
2. **Lone Pair - Bond Pair (LP-BP) Repulsions:** These are intermediate in strength.
3. **Bond Pair - Bond Pair (BP-BP) Repulsions:** These are the weakest repulsions.

This order of repulsion is a key concept in VSEPR theory and helps predict the actual shape of molecules.

Introduction to Valence Bond Theory (VBT)

Basic Assumption of VBT

Valence Bond Theory (VBT) is a crucial concept, especially for understanding coordination compounds. Its fundamental assumption is that during bond formation, the **valence orbitals** of bonding atoms participate. More specifically, it involves the overlapping of valence and singly filled orbitals.

the participating atomic orbitals.

- **Greater the extent of overlapping, greater will be the strength of the bond.**

Types of Overlapping

There are two primary types of orbital overlapping:

1. **Head-on Overlap (Axial Overlap):** This type of overlap results in the formation of a **sigma (σ) bond**.
2. **Lateral Overlap:** This type of overlap results in the formation of a **pi (π) bond**.

Internuclear Axis

Definition of Internuclear Axis

The internuclear axis is a crucial concept for understanding different types of orbital overlapping, such as head-on and lateral overlap. It is defined as the axis that passes through the nuclei of the bonding atoms.

Characteristics of the Internuclear Axis

- **Passes through Nuclei:** This axis connects the centers of the nuclei of the two atoms that are forming a bond.
- **Reference for Overlapping:** Overlapping of atomic orbitals occurs with respect to this axis.

Choosing the Internuclear Axis

While the internuclear axis can theoretically be chosen as the x-axis, y-axis, or z-axis in a coordinate system, it is conventionally chosen as the **z-axis** for studying orbital overlaps. This convention simplifies the description of how orbitals interact.

For example, if two atoms are approaching each other to form a bond, the axis passing through their nuclei is designated as the z-axis. The other two axes (x and y) are perpendicular to this internuclear (z) axis. This setup is used in Cartesian coordinates to describe the orientation of atomic orbitals during bond formation.

Types of Orbitals Based on Internuclear Axis

Internuclear Axis Convention

By convention, the Z-axis is chosen as the internuclear axis in chemistry. While X or Y axes could theoretically be chosen, using the Z-axis simplifies the understanding and representation of orbital interactions across various textbooks.

overlaps, also known as axial or sigma (σ) overlaps.

- **S-orbital:**
  - S-orbitals are spherical in shape.
  - They are equally distributed along all three axes (X, Y, Z).
  - Therefore, an S-orbital is considered to be present along the Z-axis.
- **P-orbital:**
  - Among the P-orbitals (p_x, p_y, p_z), only the p_z orbital is oriented along the Z-axis.
- **D-orbital:**
  - Among the D-orbitals ($d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}$), only the d_{z^2} orbital is oriented along the Z-axis.

These three types of orbitals (S, p_z, and d_{z^2}) are the only ones that can participate in head-on (axial) overlapping along the internuclear axis. This type of overlap leads to the formation of sigma (σ) bonds.

Orbitals Perpendicular to the Internuclear Axis

All other orbitals (e.g., $p_x, p_y, d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}$) are oriented perpendicular to the internuclear axis (Z-axis). These orbitals participate in lateral (sideways) overlapping, which leads to the formation of pi (π) bonds.

F-orbitals

F-orbitals are generally not studied in detail at this level (JEE syllabus) due to their highly complicated shapes and their deep burial within the atom, which limits their involvement in bond formation.

Head-on Overlap and Sigma Bonds

Definition of Head-on Overlap

Head-on overlap refers to the overlapping of atomic orbitals along the internuclear axis. This type of overlap is crucial for the formation of sigma (σ) bonds.

Orbitals Involved in Head-on Overlap

Only specific orbitals are oriented along the internuclear axis (typically considered the Z-axis) and can participate in head-on overlap. These include:

- S orbital
- PZ orbital

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Head-on overlaps are also known as sigma overlaps because the bond formed from this type of overlap is called a sigma (σ) bond. All sigma bonds are formed through head-on overlap.

Types of Head-on Overlap

Let's consider the internuclear axis as the Z-axis.

1. **SS Overlap:**
    - This occurs when two S orbitals overlap head-on. Both S orbitals are spherical and can overlap along the internuclear axis.
    - Example: An S orbital from one atom overlaps with an S orbital from another atom.
    - This results in the formation of a sigma bond.
2. **SP Overlap:**
    - This occurs when an S orbital overlaps head-on with a P orbital. Since the internuclear axis is the Z-axis, the P orbital involved must be the PZ orbital.
    - Example: An S orbital from one atom overlaps with a PZ orbital from another atom.
    - This also results in the formation of a sigma bond.
3. **PP Overlap:**
    - This occurs when two P orbitals overlap head-on. Again, for head-on overlap along the Z-axis, both P orbitals must be PZ orbitals.
    - Example: A PZ orbital from one atom overlaps with a PZ orbital from another atom.
    - This type of overlap also forms a sigma bond.

Key Takeaway

Regardless of whether it's SS, SP, or PP head-on overlap, the result is always the formation of a sigma (σ) bond. It is essential to remember that head-on overlap is synonymous with sigma bond formation.

Overlapping Orbitals and Internuclear Axis

For orbitals to overlap along the internuclear axis, they must be present along that axis. This type of overlap is called **head-on overlap** and results in the formation of a **sigma (σ) bond**.

Extent of Overlapping

- **p-p overlap** > **s-p overlap** > **s-s overlap**

This order is due to the surface area of overlap. p-orbitals, with their directional lobes, can point directly towards each other, leading to a greater overlap area. s-orbitals are spherical and non-directional, resulting in less effective overlap when interacting with other orbitals or with each other.

Formation of a Hydrogen Molecule (H_2)

1. **Electronic Configuration:** Each hydrogen atom has an electronic configuration of $1s^1$.
2. **Orbital Overlap:** When two hydrogen atoms approach each other, their $1s$ orbitals overlap. Overlapping means that the electron clouds of the two atoms merge, creating a common volume where electrons can be found.
3. **Electron Sharing:** Each $1s$ orbital contains one electron. Upon overlap, these two electrons are shared between the two hydrogen atoms, effectively giving each hydrogen atom a stable duplet (two electrons).
4. **Covalent Bond:** This sharing of electrons through orbital overlap forms a **covalent bond**. Since the overlap occurs along the internuclear axis (head-on), it is specifically a **sigma (σ) bond**.

Formation of a Fluorine Molecule (F_2)

1. **Electronic Configuration:** Each fluorine atom has an electronic configuration of $1s^2 2s^2 2p^5$.
2. **Participating Orbitals:** Inner (filled) orbitals ($1s^2$, $2s^2$) do not participate in bond formation. The valence $2p$ orbitals are involved.
3. **Valence Shell Configuration:** The $2p$ subshell has five electrons. According to Hund's rule, the electrons are distributed as follows:
   - $2p_x^2$
   - $2p_y^2$
   - $2p_z^1$

   The $2p_z$ orbital has one unpaired electron.
4. **Orbital Overlap:** When two fluorine atoms approach, their $2p_z$ orbitals (each containing one unpaired electron) overlap head-on along the internuclear axis (assumed to be the z-axis).
5. **Covalent Bond:** This head-on overlap of two p orbitals forms a **sigma (σ) bond**. This is specifically referred to as a **p-p overlap**.

Extent of Overlapping in Covalent Bonding

Directional Nature of Covalent Bonds

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Directionality of Orbitals

- **S-orbital:** The s-orbital is spherical and symmetrical, making it non-directional. It has electron density distributed uniformly around the nucleus.
- **P-orbital:** P-orbitals are directional. For example, a P_z orbital has electron density concentrated along the Z-axis.

Overlapping Preference

When forming a bond along a specific axis (e.g., Z-axis):

- A P_z orbital is preferred over an S-orbital because the entire electron density of the P_z orbital is aligned along the Z-axis, leading to more effective overlap.
- In contrast, an S-orbital would only have a fraction (approximately one-third) of its electron density along a specific axis, resulting in less effective overlap.

General Order of Overlapping Extent

The extent of overlapping for covalent bonding generally follows this order: PP overlap > SP overlap > SS overlap

This means that the overlap between two P-orbitals is generally more effective than the overlap between an S-orbital and a P-orbital, which in turn is more effective than the overlap between two S-orbitals. This is primarily due to the directional nature of P-orbitals, allowing for greater surface area inclusion during bond formation.

Overlapping of Orbitals: d-orbital Involvement

d-orbital Overlap with s-orbital

While discussing orbital overlaps, we consider the involvement of d-orbitals, specifically the d_{z^2} orbital. In a given shell, the energy order of orbitals is generally $s < p < d$. For example, in the fourth shell, we have $4s$, $4p$, and $4d$. This indicates that d-orbitals are relatively high in energy.

The d_{z^2} orbital can, in rare cases, overlap with an s-orbital. This is uncommon because p-orbitals exist between s and d orbitals in terms of energy, making direct s-d overlap less favorable.

d_{z^2} and p_z Overlap

Consider an overlap between a p_z orbital from one atom and a d_{z^2} orbital from another atom. The d_{z^2} orbital has a unique shape, characterized by two lobes along the z-axis and a donut-shaped ring in the xy-plane. When p_z and d_{z^2} orbitals overlap along the internuclear axis, they can form a sigma (σ) bond.

$p - p$ overlap. This is primarily due to two reasons:

1. **Size of d-orbitals:** d-orbitals are typically larger than p-orbitals, leading to less effective overlap.
2. **Higher Energy:** d-orbitals are higher in energy, which contributes to the formation of weaker bonds.

Occurrence in Molecules

Pure orbital overlapping (involving unhybridized orbitals) is predominantly observed in diatomic species. For instance, examples like H_2 and F_2 involve pure orbital overlaps. However, d-orbitals are rarely involved in diatomic molecules. This is because d-orbitals are characteristic of transition elements, and transition elements typically do not form simple diatomic molecules. Diatomic molecules are more commonly formed by elements from Group 17 and Group 18, as studied in Molecular Orbital Theory (MOT).

In polyatomic molecules, hybridization usually occurs before such pure d-orbital overlaps. Therefore, while theoretically possible, $p_z - d_{z^2}$ overlap forming a sigma bond is a rare occurrence in practical chemistry. If such an overlap were to occur, it would result in a sigma bond, but it would be weaker than a typical $p - p$ sigma bond.

Types of Overlaps: Positive and Negative

Beyond the specific orbitals involved, overlaps can be categorized into two main types:

1. **Positive Overlap (Additive or Constructive Overlap):** This type of overlap occurs when the interacting orbitals have the same phase, leading to an increase in electron density between the nuclei and the formation of a bonding molecular orbital.
2. **Negative Overlap (Subtractive or Destructive Overlap):** This occurs when orbitals of opposite phases interact, leading to a decrease in electron density between the nuclei and the formation of an antibonding molecular orbital.

Introduction to Orbital Shapes and Wave Function (Psi)

Understanding Orbital Signs

Orbitals possess signs (positive and negative) in their lobes, which represent the value of the wave function, ψ. These signs are crucial for understanding the behavior of electrons within orbitals.

S-Orbital Shape and Characteristics

- The s-orbital is spherical in shape.
- It has only one sign throughout its entire volume, typically represented as positive. This means the value of ψ is consistently positive across the entire sphere.

The positive sign does not indicate a positive charge (as electrons are negatively charged), but rather that ψ has a single, positive value throughout the orbital.

- The wave function ψ represents an electron in an orbital. If ψ is positive everywhere, it implies that the electron's presence does not involve a change in the sign of ψ within the orbital.

Nucleus Position in S-Orbital

- In an s-orbital, the nucleus is located at the center of the sphere.

P-Orbital Shape and Nodal Plane

P-Orbital Shape and Signs

- The p-orbital has a dumbbell shape, consisting of two lobes joined at the center.
- Unlike the s-orbital, the p-orbital has two opposite signs in its lobes. If one lobe is positive, the other must be negative.
- The signs (positive and negative) indicate that the value of ψ changes across the orbital.

Nucleus Position in P-Orbital

- In a p-orbital, the nucleus is located at the point where the two lobes meet, at the center of the dumbbell.

Nodal Plane in P-Orbital

- **Definition:** A nodal plane is a plane where the probability of finding an electron is zero. Mathematically, this means $\psi = 0$ on this plane.
- **Location:** In a p-orbital, the nodal plane passes through the nucleus, perpendicular to the axis connecting the two lobes. It effectively separates the positive and negative lobes.
- **Significance:** Since ψ changes from positive to negative (or vice versa) across the nodal plane, it must pass through zero at the plane itself. This is a fundamental mathematical concept: to go from a positive value to a negative value, one must cross zero.
- **Absence in S-Orbital:** An s-orbital does not have a nodal plane because ψ does not change its sign and therefore never becomes zero within the orbital. If ψ were to change its sign, it would have to pass through zero, indicating a nodal plane, which is not present in an s-orbital.

Understanding Signs in Orbitals and Nodal Planes

Signs in Orbitals

- **S orbital:** Has one sign (either positive or negative).
- **P orbital:** Has two signs (one positive, one negative). This indicates the presence of a nodal plane between the lobes.

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Nodal Plane

- A **nodal plane** is a plane where the probability of finding an electron is zero.
- Mathematically, this means the wave function (ψ) is zero, or the probability density (ψ^2) is zero, at any point on this plane.
- In orbitals, the presence of a nodal plane signifies a change in the sign of the wave function across that plane.

Significance of the Wave Function (ψ)

- ψ is a mathematical function that describes the behavior of an electron in an atom (quantum mechanical model).
- If ψ has a non-zero value at a point, it means an electron exists at that point.
- If ψ is zero at a point (e.g., on a nodal plane), it means an electron does not exist there.
- The signs (positive or negative) of ψ in different regions of an orbital do not indicate electron density. Both positive and negative regions represent equal electron density.
- The change in sign across a region simply indicates the presence of a node (a point or plane where $\psi = 0$) between those regions. For example, if an orbital goes from a positive region to a negative region, there must be a nodal plane in between where $\psi = 0$.

Summary of Orbital Signs and Nodes

- **S orbital:** One sign, no nodal planes.
- **P orbital:** Two signs (positive and negative), one nodal plane.
- **D orbital:** Opposite lobes have the same sign, indicating nodal planes between adjacent lobes where the sign changes.

Positive and Negative Overlap in Orbital Theory

Introduction to Overlap

In the context of orbital theory, the overlap between atomic orbitals is crucial for understanding chemical bonding. This overlap can be categorized into positive and negative types, each with distinct implications for bond formation.

Positive Overlap

Positive overlap occurs when atomic orbitals interact through lobes of the **same sign**. This is a mathematical concept where, if we consider the wave function (ψ) of the orbitals, lobes with the same mathematical sign (e.g., positive with positive, or negative with negative) interact constructively.

- **Electron Density:** An increase in the value of ψ directly leads to an increase in ψ^2. Since ψ^2 represents the probability of finding an electron (or electron density), positive overlap results in an **increased electron density** between the two interacting atoms.
- **Bonding Situation:** This increased electron density between atoms is the fundamental characteristic of a covalent bond. Therefore, positive overlap leads to a **bonding situation** and facilitates bond formation. According to Valence Bond Theory (VBT), only positive overlap results in the formation of a chemical bond.

Negative Overlap

Negative overlap occurs when atomic orbitals interact through lobes of **different signs**. For example, a positive lobe meeting a negative lobe.

- **Mechanism:** When lobes of different signs interact, their wave functions tend to cancel each other out. If $\psi_1 = +1$ and $\psi_2 = -1$ overlap, the resultant wave function in the overlapping region becomes $\psi_{total} = 0$. This indicates a decrease or cancellation of the wave function.
- **Nodal Plane:** Wherever the sign changes between the interacting orbitals, a **nodal plane** is formed. A nodal plane is a region where the probability of finding an electron is zero.
- **Anti-bonding Situation:** Due to the cancellation of wave functions and the formation of nodal planes, negative overlap leads to a **decrease in electron density** between the atoms. This is an **anti-bonding situation**, which is unfavorable for bond formation. In Molecular Orbital Theory (MOT), this scenario corresponds to the formation of an anti-bonding molecular orbital.

Negative Overlap and Anti-bonding Situation

Understanding Negative Overlap

Negative overlap occurs when two atomic orbitals with opposite signs (e.g., a positive s-orbital and a negative s-orbital, or a positive s-orbital and a negative lobe of a p-orbital) interact. When these orbitals meet, their wave functions cancel each other out in the overlapping region.

Consequences of Negative Overlap

- **Zero Electron Density:** In the region where negative overlap occurs, the value of the wave function (ψ) becomes zero. This means there is no electron density in between the two atoms.
- **Repulsion:** The absence of electrons between the nuclei leads to increased repulsion between the positively charged nuclei. This is contrary to the goal of a covalent bond, where electrons are shared between atoms to create an attractive force.
- **Anti-bonding Situation:** Negative overlap results in an anti-bonding situation. This means that instead of forming a stable bond, the interaction destabilizes the system and increases the repulsions between the atoms. It is the exact opposite of a bonding scenario.

overlap, their wave functions cancel out in the region of overlap, leading to zero electron density and repulsion.

Example: s-p Negative Overlap

Consider an s-orbital with a positive sign overlapping with a p-orbital where the lobe facing the s-orbital has a negative sign. In the overlapping region, the positive and negative wave functions cancel, resulting in zero electron density and repulsion. The cancellation primarily occurs in the overlapping region, not the entire orbital.

Key Takeaway

Negative overlap does not lead to bond formation; instead, it leads to repulsions between atoms, creating an unstable, anti-bonding situation. This is because the electron density, which would normally attract the nuclei, is absent in the internuclear region.

Pi Bond Formation: Lateral Overlap

Introduction to Pi Bonds

- A pi (π) bond is formed through lateral (sideways) overlap of atomic orbitals.
- This type of overlap is also referred to as pi overlap.

Understanding Lateral Overlap with p-orbitals

- Consider the internuclear axis.
- A p_x orbital is oriented perpendicular to the internuclear axis (if the internuclear axis is along the z-axis, for example).
- **Head-on overlap is not possible** for a p_x orbital along the internuclear axis because it is not aligned with it.
- Therefore, the p_x orbital from one atom will overlap sideways with a p_x orbital from another atom.
- This sideways overlap results in the formation of a pi bond.

Characteristics of Pi Bonds

- **Poor Overlap:** The sideways overlap in a pi bond is generally poor compared to the head-on overlap in a sigma bond. Orbitals cannot bend their lobes to achieve better overlap.
- **Weaker than Sigma Bonds:** Due to the poor overlap, pi bonds are inherently weaker than sigma (σ) bonds.
- **Cannot Exist Independently:** According to Valence Bond Theory (VBT), a pi bond cannot exist alone. It always requires a sigma bond to be present alongside it to strengthen the overall

Notes

Introduction to the Session

The lecturer initiated an interactive session, encouraging students to participate actively and ask doubts. The chat feature was mentioned as private, ensuring that only the lecturer could see the questions, promoting a comfortable environment for students to express their queries without hesitation. The smaller batch size was highlighted as an advantage for better interaction.

Question Setup

The session then transitioned into a question-and-answer format, starting with a question from a previous JEE Advanced exam. The initial question was framed as "Which of these is not correctly matched?" However, the lecturer later changed the question to "Which of the following is correctly matched?" to avoid confusion due to multiple 'not correctly matched' options.

Understanding Orbital Signs

During the introduction of the question, the concept of colored and uncolored lobes in orbital diagrams was discussed. This representation signifies the signs (positive or negative) of the wave function in different regions of an orbital. For instance, if a colored lobe represents a positive sign, then an uncolored lobe would represent a negative sign, and vice-versa. This concept is crucial for understanding bonding and molecular orbital theory.

Quiz Format

The question was presented in a quiz format, a common practice in the class, often utilizing Previous Year Questions (PYQs) to make the learning process engaging and fun. The timer was started, and students were encouraged to provide their answers.

Understanding Bonding and Anti-bonding Situations

Anti-bonding Situation

An anti-bonding situation occurs when lobes of opposite signs meet. For example, if a filled lobe has a positive sign and an empty lobe has a negative sign, their interaction leads to an anti-bonding state. This is characterized by a decrease in electron density between the nuclei, leading to repulsion.

Sigma Anti-bonding

In sigma anti-bonding, if a filled lobe is positive and it meets a negative lobe, it results in a sigma anti-bonding interaction. This means that the positive and negative signs of the interacting orbitals are not conducive to stable bond formation.

instance, the d_{z^2} orbital has a specific shape with two main lobes along the z-axis and a donut-shaped ring in the xy-plane. In d-orbitals, opposite lobes typically have the same sign. If the coloring or phase assignment of the lobes is incorrect (e.g., two main lobes having opposite signs when they should be the same, or the ring having an incorrect sign relative to the lobes), then the representation of the orbital is fundamentally flawed, leading to incorrect conclusions about bonding interactions.

Zero Overlap Situation

Definition of Zero Overlap

A zero overlap situation occurs when the positive and negative overlaps between interacting orbitals cancel each other out, resulting in no net bonding or anti-bonding interaction. This means that the effects of electron density increase (attraction) and electron density decrease (repulsion) are perfectly balanced.

Example: s-orbital and d-orbital Interaction

Consider an s-orbital (always positive in phase) interacting with a d-orbital. If the s-orbital overlaps with a positive lobe of the d-orbital in one region, it leads to a positive-positive overlap, which increases electron density and promotes attraction. However, if simultaneously, the s-orbital also overlaps with a negative lobe of the d-orbital in another region, it leads to a positive-negative overlap, which decreases electron density and promotes repulsion.

Did you find these notes helpful?

If these two effects (attra n from positive-negative overlap) are equal in mag it is zero overlap, meaning no net change in electron density between the atoms, and thus no bond formation or destabilization. This is a common scenario in certain orbital interactions where symmetry dictates the cancellation of bonding and anti-bonding contributions.

Understanding d-orbitals and Signs of Lobes

d-orbital Lobes and Signs

In d-orbitals, opposite lobes generally have the same sign. However, the d_{z^2} orbital is unique in its shape and sign distribution.

- d_{z^2} **Orbital:** This orbital has two main lobes along the z-axis, which share the same sign (either positive or negative). It also features a toroidal (doughnut-shaped) ring around the xy-plane, which has the opposite sign to the lobes. For example, if the lobes are positive, the ring is negative, and vice-versa.

Rules for Drawing Molecular Structures

Notes

1. **Select the Central Atom:**
    - The central atom is typically the **least electronegative atom** in the molecule.
    - **Exception:** Hydrogen can never be a central atom because it forms only one bond. A central atom must be capable of forming multiple bonds to connect other atoms.
2. **Identify Side/Surrounding Atoms:**
    - Once the central atom is selected, all other atoms in the molecule are considered side or surrounding atoms.
3. **Complete Valencies of Side Atoms:**
    - Side atoms have a **fixed valency**, meaning they typically form a specific number of bonds (e.g., halogens form one bond, oxygen forms two bonds).
    - Take advantage of this fixed valency to complete the bonding requirements of the side atoms first.
4. **Determine Valency of Central Atom:**
    - The central atom can exhibit **variable valency**, meaning the number of bonds it forms can change depending on the molecule (e.g., it might form three, five, or six bonds).
    - After satisfying the valencies of the side atoms, the remaining bonding capacity will be fulfilled by the central atom, which will adjust its valency accordingly.

Types of Side Atoms and Their Bonding Characteristics

Introduction to Side Atoms

Side atoms play a crucial role in determining the structure and properties of molecules. Understanding their bonding behavior is fundamental in chemistry. This section will cover the bonding characteristics of common side atoms: Hydrogen, Halogens, Oxygen, and Nitrogen.

1. Hydrogen Atom

- **Valence Electrons:** Hydrogen has only one electron.
- **Bonding:** It always forms **one bond**.
- **Lone Pairs:** Due to having only one electron, it forms no lone pairs.

2. Halogen Atoms (F, Cl, Br, I)

- **Valence Electrons:** Halogens (Fluorine, Chlorine, Bromine, Iodine) always have **seven valence electrons**.

- **Bonding:** They form only one bond.
- **Lone Pairs:** They will have **three lone pairs**.

3. Oxygen Atom

- **Valence Electrons:** Oxygen has **six valence electrons**.
- **Octet Completion:** It needs two more electrons to complete its octet.
- **Bonding:** It forms **two bonds**. These can be two single bonds or one double bond.
- **Lone Pairs:** It will have **two lone pairs**.
  - **Example (Water):** In a water molecule (H_2O), oxygen forms two single bonds with two hydrogen atoms and has two lone pairs.
  - **Example (Oxygen Molecule):** In an oxygen molecule (O_2), each oxygen atom shares two electrons with the other oxygen atom, forming a double bond. Each oxygen atom also retains two lone pairs.

4. Nitrogen Atom

- **Valence Electrons:** Nitrogen has **five valence electrons**.
- **Octet Completion:** It needs three more electrons to complete its octet.
- **Bonding:** It forms **three bonds**.
- **Lone Pairs:** It will have **one lone pair**.

Practice Question: Lone Pairs on Central Atoms

Question

Calculate the total number of lone pairs on the central atom in the following species: SNF_3, SO_2, and CO_2.

Options

A. 1 B. 2 C. 3 D. 0

Solution Approach

To solve this, we need to determine the number of lone pairs on the central atom for each molecule individually and then sum them up.

1. **SNF_3 (Thionyl Fluoride)**
   - Central atom: Sulfur (S)
   - Valence electrons of S: 6

lecturer meant a sulfur-containing compound with fluorine and nitrogen, but the most common interpretation for SNF3 would be N as central atom. Let's re-evaluate based on common stable molecules. If it's SNF_3, then Nitrogen is the central atom. Let's proceed with Nitrogen as the central atom for SNF_3.)

- **Re-evaluation for SNF_3 (Nitrogen as central atom):**
  - Central atom: Nitrogen (N)
  - Valence electrons of N: 5
  - Number of bonds: N forms three single bonds with F atoms.
  - Electrons used in bonding: $3 \times 1 = 3$
  - Remaining electrons: $5 - 3 = 2$
  - Number of lone pairs: $2/2 = 1$
  - So, SNF_3 has 1 lone pair on the central N atom.

2. **SO_2 (Sulfur Dioxide)**

- Central atom: Sulfur (S)
- Valence electrons of S: 6
- Number of bonds: S forms two double bonds with O atoms (or one double and one single with resonance, but for lone pair calculation, consider the number of atoms bonded).
- Electrons used in bonding: S forms bonds with two oxygen atoms. In SO_2, sulfur forms one double bond and one single bond (with resonance), or two double bonds if we consider formal charge. Let's use the VSEPR approach. Sulfur is bonded to two oxygen atoms. To satisfy octets, S will have one lone pair.
- Alternatively, total valence electrons = $6(S) + 2 \times 6(O) = 18$
- S-O single bonds: $2 \times 2 = 4$ electrons
- Remaining electrons for octets: $18 - 4 = 14$
- Place 6 electrons on each O: $2 \times 6 = 12$ electrons
- Remaining electrons: $14 - 12 = 2$ electrons
- These 2 electrons go to the central S atom as a lone pair.
- Number of lone pairs on S: $2/2 = 1$

3. **CO_2 (Carbon Dioxide)**

- Central atom: Carbon (C)
- Valence electrons of C: 4
- Number of bonds: C forms two double bonds with O atoms.

- Electrons used in bonding: 4 − 2 = 2 (if considering double bonds)
- Electrons used in bonding (for lone pair calculation): C forms 4 bonds in total (two double bonds). All 4 valence electrons are used in bonding.
- Remaining electrons: 4 − 4 = 0
- Number of lone pairs: 0/2 = 0

Total Lone Pairs

Total lone pairs = (Lone pairs on N in SNF_3) + (Lone pairs on S in SO_2) + (Lone pairs on C in CO_2)
Total lone pairs = 1 + 1 + 0 = 2

Therefore, the total number of lone pairs on the central atoms is 2.

Final Answer

The correct option is B.

Approach to Drawing Lewis Structures

Identifying the Central Atom

- The central atom is typically the least electronegative atom in the molecule, excluding fluorine.
- Fluorine (F) is almost never the central atom due to its high electronegativity.

Completing Valency of Side Atoms

- Always complete the valency of the side atoms first, not the central atom.
- **Fluorine (F):** Forms one bond with the central atom.
- **Nitrogen (N):** Forms three bonds to complete its valency.
- **Oxygen (O):** Forms two bonds to complete its valency.

Drawing Lewis Structure of SNF_3

Step-by-Step Construction

1. **Identify Central Atom:** Sulfur (S) is the central atom.
2. **Side Atoms:** Nitrogen (N) and Fluorine (F) are side atoms.
3. **Bonds with Side Atoms:**
   - Each Fluorine atom forms one bond with Sulfur.
   - The Nitrogen atom forms three bonds with Sulfur.
4. **Valence Electrons of Sulfur:** Sulfur has 6 valence electrons.

6. **Lone Pairs on Central Atom:** Since all 5 valence electrons are shared, there are no lone pairs on the central Sulfur atom in SNF_3.

Structure of SNF_3

F
$S = N$
F

Drawing Lewis Structure of SO_2

Step-by-Step Construction

1. **Identify Central Atom:** Sulfur (S) is the central atom.
2. **Side Atoms:** Oxygen (O) atoms are side atoms.
3. **Bonds with Side Atoms:** Each Oxygen atom forms two bonds with Sulfur.
4. **Valence Electrons of Sulfur:** Sulfur has 6 valence electrons.
5. **Electron Sharing:** Sulfur forms a total of 4 bonds (two with each Oxygen). This means 4 valence electrons of Sulfur are shared.
6. **Lone Pairs on Central Atom:** Out of 6 valence electrons, 4 are shared, leaving $6 - 4 = 2$ unshared electrons. These 2 unshared electrons form one lone pair on the central Sulfur atom.

Structure of SO_2

$O = S = O$

Drawing Lewis Structure of CO_2

Step-by-Step Construction

1. **Identify Central Atom:** Carbon (C) is the central atom.
2. **Side Atoms:** Oxygen (O) atoms are side atoms.
3. **Bonds with Side Atoms:** Each Oxygen atom forms two bonds with Carbon.
4. **Valence Electrons of Carbon:** Carbon has 4 valence electrons.
5. **Electron Sharing:** Carbon forms a total of 4 bonds (two with each Oxygen). This means all 4 valence electrons of Carbon are shared.
6. **Lone Pairs on Central Atom:** Since all 4 valence electrons are shared, there are no lone pairs on the central Carbon atom in CO_2.

Structure of CO_2

Notes

To find the total number of lone pairs on the central atoms of SNF_3, SO_2, and CO_2:

- Lone pairs on central atom in SNF_3: 0
- Lone pairs on central atom in SO_2: 1
- Lone pairs on central atom in CO_2: 0

Total lone pairs = 0 + 1 + 0 = 1.

Homework and Preparation for Next Session

Importance of Regularity and Homework

To succeed in the JEE Mains and Advanced exams, students must commit to two key practices:

- **Regular Class Attendance:** Attend all classes without fail. Even if feeling unwell, use earphones and listen to the lecture. The online format makes this more accessible.
- **Diligent Homework Completion:** Complete all homework assignments given by teachers religiously. This is crucial for reinforcing learning and preparing for exams.

Next Steps for JEE Preparation

Following these two practices will significantly increase the chances of cracking the target exams. The goal is to be well-prepared for JEE Advanced by next year, having already secured a good rank in JEE Mains.

Homework for the Next Class

Before the next session, which is scheduled for the day after tomorrow, students are required to:

1. **Revise and Make Notes:** Review the current session's content, write it down in notebooks, and create fair notes.
2. **NCERT Revision:** Revise the chapter "Chemical Bonding" from the NCERT textbook. This chapter is very important for both JEE Mains and Advanced. Read it thoroughly, as it will be enjoyable and beneficial.
3. **Electronic Configuration (Optional but Recommended):** If there are any doubts regarding writing electronic configurations, revise the configuration part of the atomic structure chapter. Understanding how to write electronic configurations is also important.

Any doubts encountered during this revision will be addressed in the next class, after which the topic of coordination chemistry will be started.

Notes